



GUJARAT TECHNOLOGICAL UNIVERSITY

Shree Swami Atmanand Saraswati Institute of Technology

A

Project Report on Food Waste Prediction and Optimization

Under subject of

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B.E 3rd year, semester – V
COMPUTER ENGINEERING

Under the Guidance of
Prof. Vipul S. Kania

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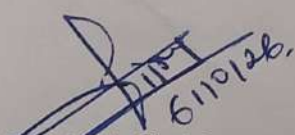


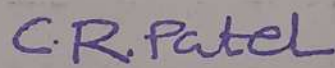
CERTIFICATE

This is to certify that the Project entitled “Food Waste Prediction and Optimization” has been prepared by Kasodariya Heet (240760107087), Khunt Deep(240760107093), Kachhadiya Krish (240760107097), Hingu Krishn(240760107098), Maniya Shubh(240760107115), Kuvadiya Om (240760107132) in the subject of Python for Data Science (BE05000231) under my guidance in partial fulfillment of the degree of Bachelor of Engineering in Computer Engineering Department (5th Semester) of Gujarat Technological University, Ahmedabad during the academic year 2026-2027.

Date:

Place: SSASIT, Surat


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1. ACKNOWLEDGEMENT

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2. ABSTRACT

Food waste is an important issue in large-scale food preparation environments such as college canteens, hostels, restaurants, and institutional kitchens. Preparing more food than required can result in unnecessary waste and increased expenditure. Therefore, an effective method is required to estimate food demand and improve food preparation decisions.

The Food Waste Prediction And Optimization is a Python-based data science project developed to analyze food consumption and waste patterns and provide data-driven recommendations. The system uses information such as students, food prepared, food consumed, food wasted, menu items, day, weather, temperature, holidays, examination days, and special events.

The project includes data preparation and validation followed by Exploratory Data Analysis (EDA) to identify patterns in food consumption and waste. Visualizations are used to analyze waste according to menu items, days, months, and weather conditions.

Machine learning techniques are used to predict food consumption. The project evaluates Linear Regression, Random Forest, and Gradient Boosting models. The prediction results are used by the Prediction Engine and Recommendation Engine to estimate the required food quantity with a suitable safety buffer.

The system also includes a Before-vs-After Simulator to compare traditional preparation with AI-assisted recommendations. A Streamlit-based application provides an interactive interface for analysis, prediction, recommendations, and simulation.

The project combines data analysis, machine learning, prediction, recommendation, and visualization for food waste management. Since the dataset is custom/synthetic, the demonstrated savings are simulation-based rather than measured real-world outcomes.

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3. INTRODUCTION

Food waste is a significant issue in places where food is prepared in large quantities, such as college canteens, hostels, restaurants, and institutional kitchens. When the amount of food prepared is higher than the actual consumption, the remaining food may become waste. This can lead to unnecessary use of food resources, increased cost, and inefficient food preparation.

Traditional food preparation decisions are often based on approximate estimates or previous experience. However, food consumption can change depending on factors such as the number of students, menu items, day of the week, weather conditions, temperature, holidays, examination days, and special events. Therefore, using these factors for data analysis can help in understanding consumption and waste patterns.

The Food Waste Prediction and Optimization project is developed using Python, data analysis, machine learning, and Streamlit. The system analyzes food consumption and waste data to identify important patterns and predict the expected food consumption for given conditions.

The project begins with data preparation and validation, followed by Exploratory Data Analysis (EDA). Machine learning models are then applied to predict food consumption. The prediction is passed to the Prediction Engine and Recommendation Engine to calculate a suitable food preparation quantity with a safety buffer. A Before-vs-After Simulator is also included to compare traditional preparation with AI-assisted recommendations.

The complete system is integrated into a Streamlit application that provides an interactive interface for analysis, prediction, recommendation, and simulation. The project demonstrates how data-driven techniques can be used to support better food preparation decisions and contribute toward food waste prediction and optimization.

4. PROBLEM STATEMENT

In large-scale food preparation environments, determining the exact quantity of food required is difficult. Preparing too much food can result in unnecessary food waste, while preparing too little food may be insufficient for the expected number of consumers.

Food consumption is influenced by several factors, including the number of students, menu, day, weather, temperature, holidays, examination days, and special events. Manually considering all these factors can be difficult and may lead to inaccurate preparation decisions.

Therefore, there is a need for a system that can:

- Analyze historical food consumption and waste data.
- Identify patterns and factors related to food waste.
- Predict the expected food consumption using machine learning.
- Calculate a suitable food preparation quantity using the prediction and a safety buffer.
- Compare traditional food preparation with AI-assisted recommendations.
- Provide an interactive interface for users to perform analysis and predictions.

The Food Waste Prediction and Optimization project addresses this problem by combining data analysis, machine learning, prediction, recommendation, simulation, and visualization into a single system.

5. OBJECTIVES

The main objectives of the Food Waste Prediction and Optimization project are:

1. To prepare and validate the dataset

To clean, validate, and preprocess the food consumption and food waste dataset before analysis.

2. To analyze food consumption and waste patterns

To perform Exploratory Data Analysis (EDA) and identify trends related to food preparation, consumption, and waste.

3. To study the effect of different factors

To analyze the relationship between food waste and factors such as menu, day, weather, temperature, holidays, examination days, and special events.

4. To develop a machine learning model

To implement and compare Linear Regression, Gradient Boosting Regression, and Random Forest Regression for predicting food consumption.

5. To develop a Prediction Engine

To predict expected food consumption based on user-provided input conditions.

6. To develop a Recommendation Engine

To calculate the recommended food preparation quantity by applying an appropriate safety buffer to the predicted consumption.

7. To develop a Before-vs-After Simulator

To compare traditional food preparation with AI-assisted recommendations and estimate potential reduction in food waste.

8. To integrate all modules into one system

To connect data preparation, EDA, machine learning, prediction, recommendation, and simulation into a complete workflow.

9. To develop an interactive Streamlit application

To provide a user-friendly interface for analyzing data, generating predictions, obtaining recommendations, and viewing simulation results.

10. To support data-driven food preparation decisions

To demonstrate how machine learning and data analysis can assist in improving food preparation planning and food waste optimization.

CHAPTER 1: DATA PREPARATION AND VALIDATION

1.1 Introduction

Data preparation is the first important step in the Food Waste Prediction and Optimization System. Before performing exploratory data analysis and machine learning, the available food-service data must be checked, cleaned, and prepared properly.

The project uses a custom dataset containing 300 records and 18 columns. Each record represents one food-service observation and contains information about the date, menu, number of students, food prepared, food consumed, food wasted, weather conditions, temperature, and special conditions.

The purpose of data preparation is to ensure that the dataset is consistent and suitable for further analysis and machine learning.

1.2 Dataset Collection and Structure

A custom dataset was created for the academic demonstration of the Food Waste Prediction and Optimization System. The dataset contains 300 records and 18 columns.

The dataset includes operational information such as:

- Date of food-service record
- Menu item
- Number of actual students
- Food prepared in portions
- Food consumed in portions
- Food wasted in portions
- Consumption rate
- Waste percentage
- Estimated waste in kilograms
- Food cost per portion
- Waste cost
- Day of the week
- Weather condition
- Temperature
- Holiday status
- Examination-day status
- Special-event status

The dataset is used as historical data for analyzing food-waste patterns and predicting food consumption.

Table 1.1: Important Dataset Information

Parameter	Value
Total Records	300
Total Columns	18
Initially Invalid Records	5
Invalid Record Percentage	1.6667%

1.3 Data Cleaning

Data cleaning was performed before continuing with exploratory data analysis and machine learning. During the initial validation, five records were identified where the value of "food_consumed_portions" was greater than "actual_students".

According to the project's data definition, these records were considered logically invalid because the number of consumed portions should not exceed the recorded number of students.

The invalid consumed values were corrected to the corresponding number of actual students. After correction, all dependent fields were recalculated to maintain consistency within the dataset.

1.4 Data Validation

After cleaning the dataset, the validation process was performed again to check whether any logically invalid records remained.

The validation rule used was:

food_consumed_portions > actual_students

After correction, the validation query returned an empty DataFrame, which confirmed that no records violated this particular constraint.

This validation step ensured that the cleaned dataset was ready for further analysis and machine learning.

1.5 Data Preprocessing

After correcting the invalid records, the dependent measures were recalculated using the project's defined formulas.

The main calculations were:

Food Wasted Portions

Food Wasted = Food Prepared – Food Consumed

Consumption Rate

Consumption Rate = Food Consumed / Actual Students

Waste Percentage

Waste Percentage = (Food Wasted / Food Prepared) × 100

Estimated Food Waste

Estimated Waste (kg) = Food Wasted × 0.35

Waste Cost

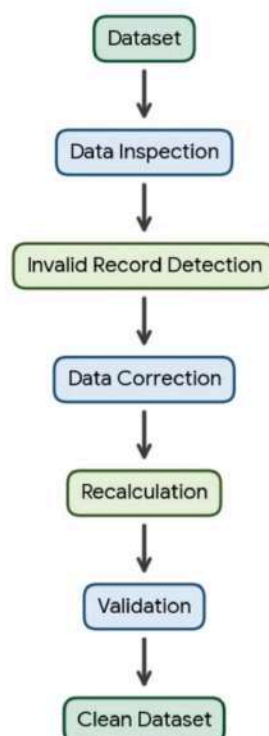
Waste Cost = Food Wasted × Food Cost per Portion

These calculations ensure that the waste-related values remain consistent with the corrected food-consumption data.

1.6 Data Preparation Workflow

The data preparation process follows these steps:

1.6 Data Preparation Workflow



The cleaned dataset is then passed to the next stage of the project, which is Exploratory Data Analysis (EDA).

1.7 Chapter Summary

In this chapter, the dataset used for the Food Waste Prediction and Optimization System was introduced and prepared for further processing. The dataset contains 300 records and 18 columns. Initially, five logically invalid records were identified and corrected.

After correcting the records, dependent values such as food waste, consumption rate, waste percentage, estimated waste weight, and waste cost were recalculated. A final validation confirmed that the identified logical constraint was satisfied.

The resulting cleaned dataset provides a consistent foundation for the next chapter, Exploratory Data Analysis (EDA).

CHAPTER 2: EXPLORATORY DATA ANALYSIS (EDA)

2.1 Introduction

Exploratory Data Analysis (EDA) is performed to understand the important patterns, relationships, and trends present in the Food Waste Prediction and Optimization dataset.

The analysis focuses on food preparation, food consumption, food waste, menu items, days, weather conditions, and special conditions. EDA helps in understanding the behavior of the dataset before applying machine learning models.

The dataset contains 300 records. The main overall results obtained from the dataset are shown below.

Table 2.1: Overall Food Waste Statistics

Metric	Result
Total Records	300
Average Food Prepared	403.07 portions
Average Food Consumed	340.32 portions
Average Food Wasted	62.75 portions
Average Waste Percentage	15.83%
Total Food Wasted	18,824 portions

2.2 Food Prepared, Consumed and Wasted

The first EDA analysis compares the amount of food prepared, consumed, and wasted.

On average, 403.07 portions of food were prepared, while 340.32 portions were consumed. The average food waste was 62.75 portions.

This comparison helps identify the difference between food preparation and actual consumption. It also provides a basic understanding of the amount of food that may remain unused.

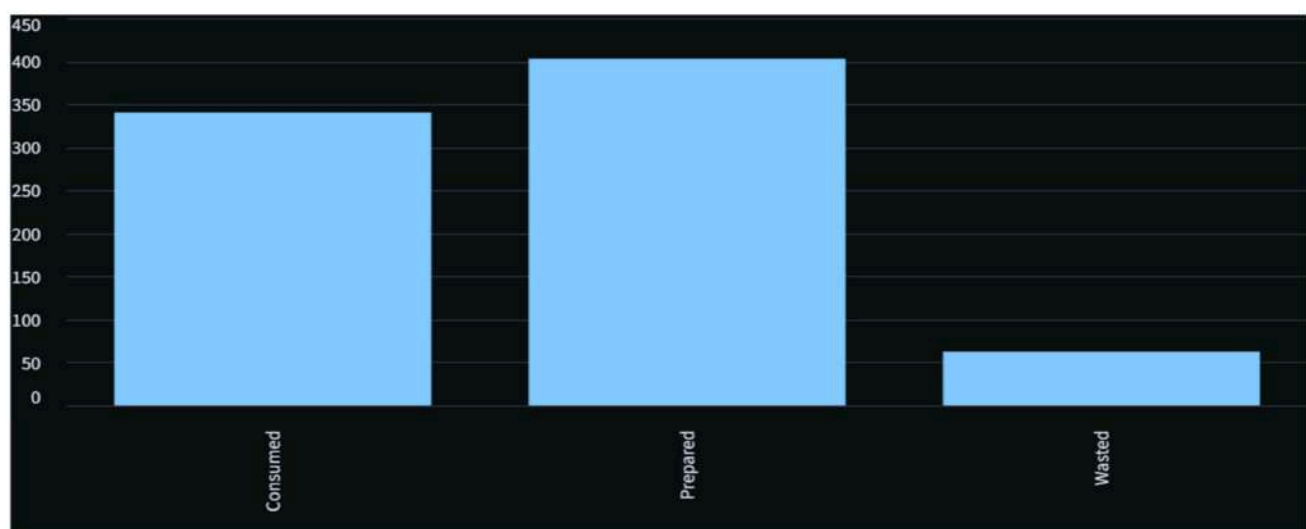


Figure 2.1: Comparison of food prepared, consumed, and wasted portions.

Figure Description:

This graph compares the average quantity of food prepared, consumed, and wasted. It shows that the prepared quantity is higher than the consumed quantity, resulting in a measurable amount of food waste.

2.3 Waste Percentage by Menu

Different menu items may have different consumption and waste patterns. Therefore, waste percentage was analyzed for different menu items in the dataset.

The menu-wise analysis helps identify food items that show relatively higher or lower waste percentages. This information can be useful for making menu-specific preparation decisions.

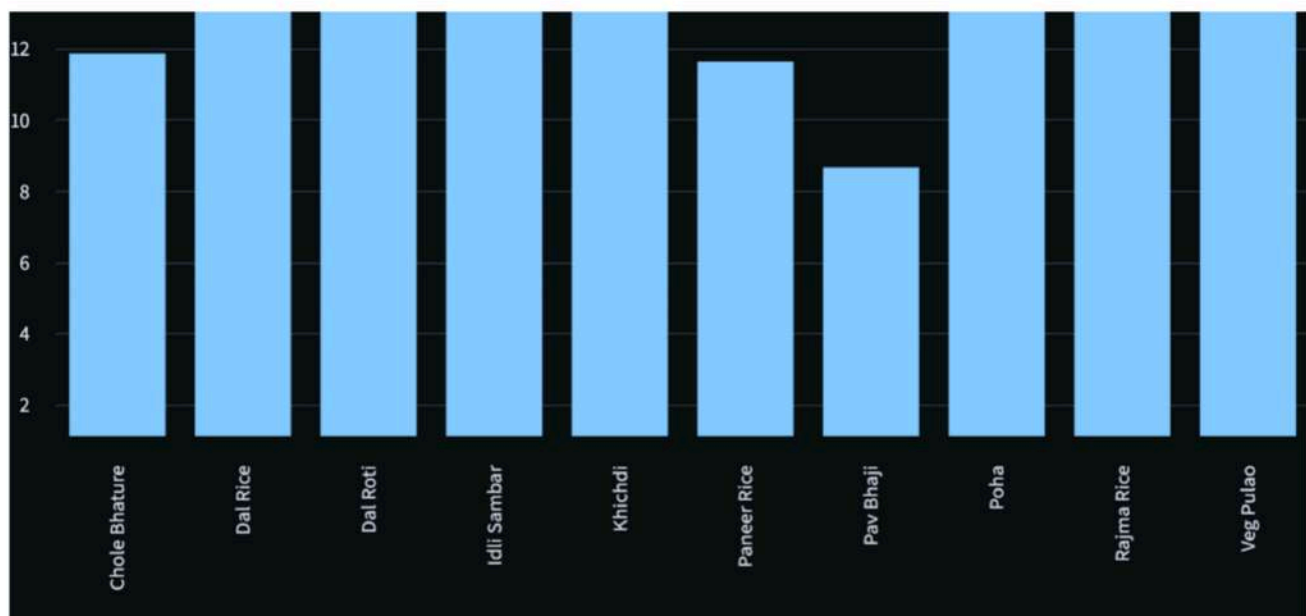


Figure 2.2: Waste percentage comparison across different menu items.

Figure Description:

This graph compares the percentage of food wasted for different menu items. It helps identify variations in waste behavior among the available food items.

2.4 Waste Percentage by Day

Food consumption can vary depending on the day of the week. To understand this variation, the waste percentage was analyzed across different days.

This analysis provides information about whether certain days show different waste behavior compared with other days.

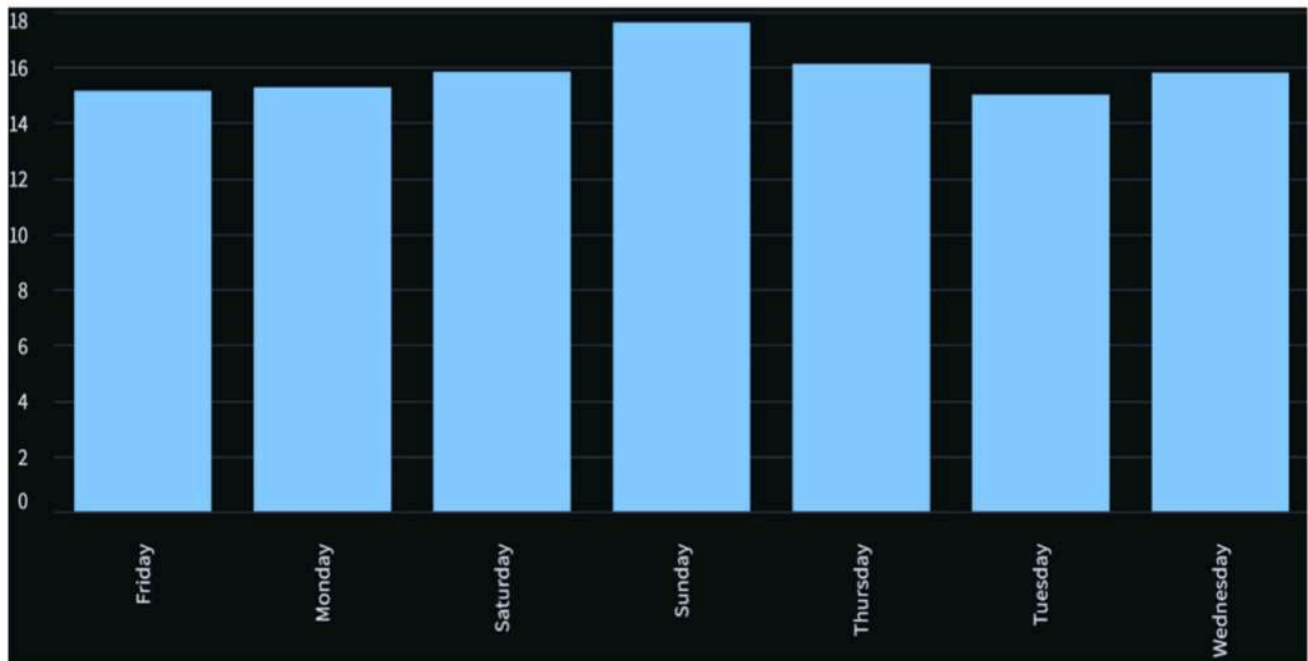


Figure 2.3: Food-waste percentage across different days of the week.

Figure Description:

This graph shows the waste percentage for different days. The comparison helps identify daily variations in food-waste behavior.

2.5 Monthly Food Waste Trend

A monthly analysis was performed to observe how estimated food waste changes over time.

The monthly trend provides a visual representation of food-waste behavior across the recorded period. Such trends can help in identifying periods where waste levels may require additional attention.

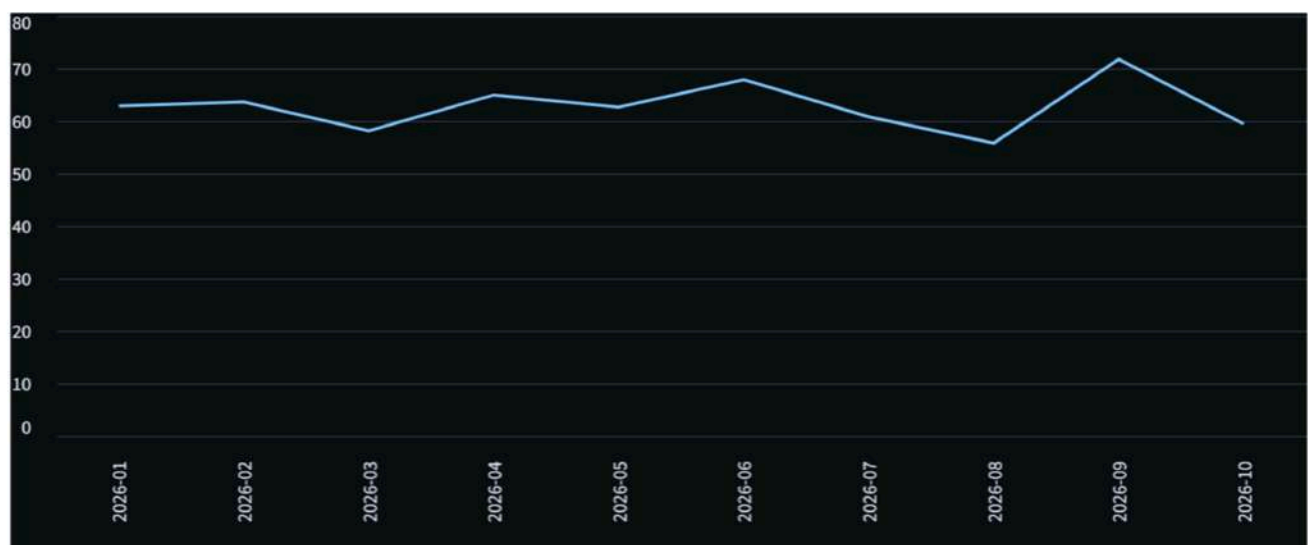


Figure 2.4: Monthly trend of estimated food waste.

Figure Description:

This graph presents the estimated food waste across different months in the dataset. It helps visualize changes in waste levels over time.

2.6 Weather-wise Food Waste Analysis

Weather conditions were also considered during the analysis because environmental conditions may be associated with changes in food consumption.

The dataset contains four weather categories: Hot, Cloudy, Normal, and Rainy.

Table 2.2: Food Waste by Weather Condition

Weather	Average Students	Average Consumed	Average Wasted	Waste %
Hot	389.63	331.00	74.67	18.59%
Cloudy	373.75	331.31	59.31	16.31%
Normal	388.57	346.29	59.26	14.85%
Rainy	374.93	338.52	56.36	14.60%

The analysis shows differences in average waste across the four weather categories. Hot-weather records have the highest average waste percentage in this dataset, while rainy-weather records have the lowest.

These values describe the patterns in the custom dataset and should not be interpreted as proof that weather directly causes food waste.

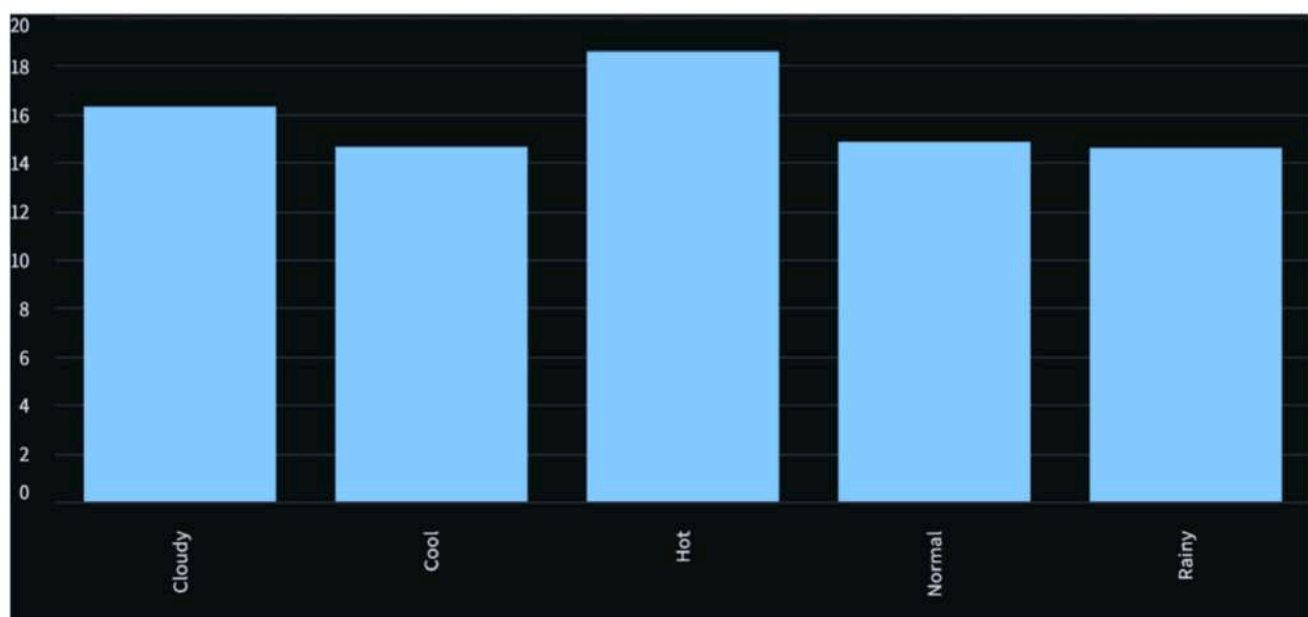


Figure 2.5: Food-waste analysis under different weather conditions.

Figure Description:

This graph compares food-waste behavior under Hot, Cloudy, Normal, and Rainy weather conditions. It helps visualize how the recorded waste percentage varies across weather categories.

2.7 Special Conditions Analysis

The dataset also includes information about holidays, examination days, and special events.

These conditions were compared to determine whether the recorded waste percentage changes under different circumstances.

Table 2.3: Waste Percentage under Special Conditions

Condition	No	Yes
Holiday	15.46%	18.31%
Exam Day	15.85%	15.74%
Special Event	15.71%	17.03%

The results show that the recorded waste percentage varies between normal and special conditions. Holiday and special-event records show higher average waste percentages in this dataset, while the difference for examination days is small.

These observations are descriptive and do not establish a causal relationship.

2.8 Correlation Analysis

Correlation analysis was used to understand the linear relationships between selected numerical variables.

Table 2.4: Selected Correlations

Variables	Correlation
Actual Students vs Food Prepared	0.99
Actual Students vs Food Consumed	0.98
Food Wasted vs Waste Percentage	0.63
Temperature vs Food Consumed	-0.04
Temperature vs Waste Percentage	0.16

The strong correlation between actual students and food prepared (0.99) indicates that preparation quantity is closely associated with the number of students.

Similarly, the correlation between actual students and food consumed (0.98) is also strong.

The temperature correlations are relatively small in this dataset, indicating that temperature alone does not explain most of the variation in consumption or waste.

Correlation represents association and does not prove that one variable causes another.

2.9 Menu Intelligence

Menu-level analysis was performed to understand differences in food-waste behavior between menu items.

This analysis can support the recommendation system by providing additional context about menu-specific waste patterns. The information can be used along with predicted consumption when making food-preparation decisions.

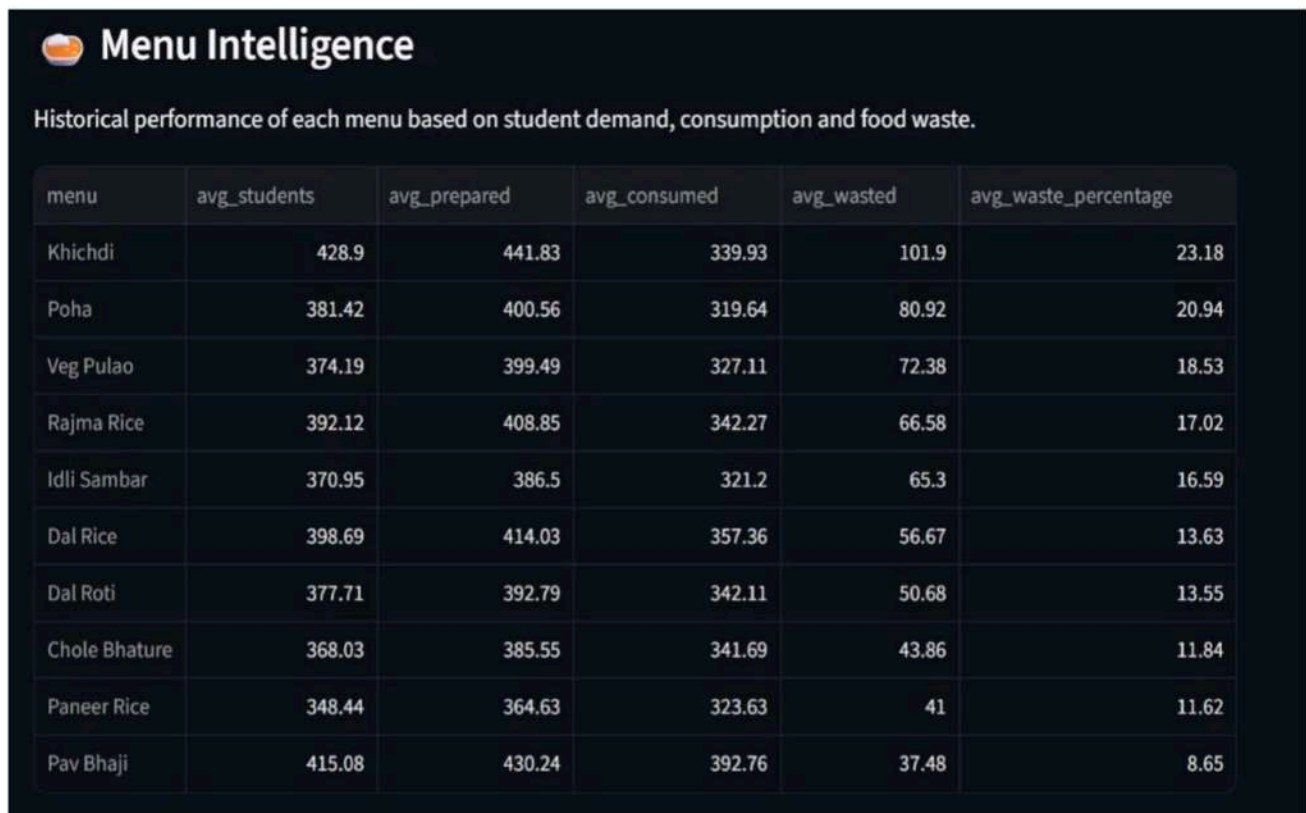


Figure 2.6: Menu-wise food-waste analysis and menu intelligence.

Figure Description:

This figure presents the menu-level analysis generated by the system. It helps compare waste behavior across different menu items and provides useful information for food-preparation planning.

2.10 EDA Summary

The EDA provided important information about the food-waste dataset.

The analysis showed that the dataset had an average preparation quantity of 403.07 portions, average consumption of 340.32 portions, and average waste of 62.75 portions. The overall average waste percentage was 15.83%.

Menu, day, weather, and special-condition analyses showed variations in recorded food-waste behavior. Correlation analysis also identified strong relationships between the number of students and both food preparation and consumption.

The findings from EDA provide a foundation for the next stage of the project, where machine learning models are used to predict food consumption.

2.11 Chapter Summary

In this chapter, Exploratory Data Analysis was performed on the cleaned Food Waste Prediction and Optimization dataset. Different analyses were used to understand food preparation, consumption, waste, menu behavior, daily patterns, monthly trends, weather conditions, and special conditions.

The EDA results provide useful insights into the dataset and help identify the variables that can be considered during the machine learning stage.

CHAPTER 3: MACHINE LEARNING MODEL

3.1 Introduction

Machine Learning is an important part of the Food Waste Prediction And Optimization. It is used to predict the number of food portions that are likely to be consumed under given conditions.

The project uses a supervised regression approach because the target variable, "food_consumed_portions", is a numerical value. The model learns relationships between different operational conditions and historical food consumption.

Three regression models were evaluated:

- Linear Regression
- Gradient Boosting Regression
- Random Forest Regression

The models were trained and evaluated using the cleaned dataset.

3.2 Machine Learning Approach

The objective of the machine learning model is to predict food consumption before food preparation. The model uses operational information available before preparation as input and produces the expected food consumption in portions as output.

Input Features

The following features were used:

- Expected/actual students
- Menu
- Day of the week
- Weather
- Temperature
- Holiday status
- Examination-day status
- Special-event status

Target Variable

The target variable is:

"food_consumed_portions"

The model therefore learns the relationship between the input conditions and the number of portions expected to be consumed.

3.3 Train-Test Split

The dataset contains 300 records. It was divided into training and testing data.

- Training records: 240
- Testing records: 60

The training data was used to develop the models, while the testing data was kept separate to evaluate their prediction performance on unseen records.

This separation helps measure how well the models perform beyond the records used during training.

3.4 Linear Regression

Linear Regression was used as one of the prediction models because it provides a simple and interpretable relationship between input features and the target variable.

The model attempts to estimate food consumption based on the given operational conditions.

The recorded test results for Linear Regression were:

- MAE: 14.11 portions
- RMSE: 16.84 portions
- R^2 : 0.980

The results indicate that the model produced relatively small prediction errors on the recorded test dataset.

expected_students	menu	day_of_week	weather	temperature_c	is_holiday	is_exam_day	special_event	Actual Consumption	Predicted Consumption	Prediction Error
135	Idli Sambar	Thursday	Normal	27.9	1	0	0	119	113.79	5.21
507	Rajma Rice	Thursday	Hot	36.6	0	0	0	440	426.27	13.73
447	Dal Roti	Tuesday	Normal	30.4	0	1	0	398	401.28	-3.28
346	Idli Sambar	Saturday	Hot	36.6	0	0	0	281	280.88	0.20
364	Idli Sambar	Saturday	Normal	31.2	0	0	0	323	306.53	16.47
370	Veg Pulao	Saturday	Normal	30.8	0	0	0	270	311.89	-41.89
431	Rajma Rice	Thursday	Cool	21.0	0	1	0	375	382.06	-7.06
194	Pav Bhaji	Monday	Cloudy	30.4	1	0	0	172	191.34	-19.34
501	Dal Rice	Tuesday	Hot	33.5	0	0	0	418	433.67	-15.67
..										
488	Dal Roti	Wednesday	Normal	32.0	0	0	0	418	429.35	-11.35
505	Rajma Rice	Friday	Normal	23.7	0	0	0	431	446.04	-15.04
222	Rajma Rice	Sunday	Normal	30.4	0	0	0	185	193.43	-8.43
415	Khichdi	Monday	Rainy	31.0	0	1	0	324	334.59	-10.59

Figure 3.1: Linear Regression model evaluation output.

Description:

This screenshot shows the output of the Linear Regression model, including its prediction results and evaluation metrics. It demonstrates the performance of the model on the test data.

3.5 Gradient Boosting Regression

Gradient Boosting Regression was also evaluated as a nonlinear ensemble method. It can model more complex relationships between the input features and food consumption.

The recorded test results were:

- MAE: 16.06 portions
- RMSE: 19.53 portions
- R^2 : 0.973

The model showed strong performance on the recorded test dataset.

expected_students	menu	day_of_week	weather	temperature_c	is_holiday	is_exam_day	special_event	Actual Consumption	Predicted Consumption	Prediction Error
135	Idli Sambar	Thursday	Normal	27.9	1	0	0	119	112.88	6.12
507	Rajma Rice	Thursday	Hot	36.6	0	0	0	440	434.38	5.62
447	Dal Roti	Tuesday	Normal	30.4	0	1	0	398	407.45	-9.45
346	Idli Sambar	Saturday	Hot	36.6	0	0	0	281	267.42	13.58
364	Idli Sambar	Saturday	Normal	31.2	0	0	0	323	300.55	22.45
370	Veg Pulao	Saturday	Normal	30.8	0	0	0	270	310.11	-40.11
431	Rajma Rice	Thursday	Cool	21.0	0	1	0	375	371.78	3.22
194	Pav Bhaji	Monday	Cloudy	30.4	1	0	0	172	187.91	-15.91
501	Dal Rice	Tuesday	Hot	33.5	0	0	0	418	431.04	-13.04
481	Paneer Rice	Thursday	Cloudy	30.8	0	0	0	457	421.01	35.99
460	Poha	Wednesday	Cool	19.4	0	1	0	352	387.93	-35.93
544	Dal Rice	Friday	Normal	26.5	0	0	0	442	473.98	-31.98
523	Veg Pulao	Friday	Normal	28.7	0	0	0	442	460.81	-18.81
211	Dal Roti	Sunday	Cool	25.6	0	0	0	211	180.42	30.58
397	Chole Bhature	Thursday	Normal	30.8	0	1	0	353	369.49	-16.49
182	Dal Rice	Monday	Hot	36.3	1	0	0	164	147.21	16.79
488	Dal Roti	Wednesday	Normal	32.0	0	0	0	418	421.00	-3.00
505	Rajma Rice	Friday	Normal	23.7	0	0	0	431	441.10	-10.10
222	Rajma Rice	Sunday	Normal	30.4	0	0	0	185	188.57	-3.57
415	Khichdi	Monday	Rainy	31.0	0	1	0	324	344.06	-20.06

Figure 3.2: Gradient Boosting Regression model evaluation output.

Description:

This screenshot shows the evaluation output of the Gradient Boosting model, including its prediction error and R^2 value on the test data.

3.6 Random Forest Regression

Random Forest Regression was the third model evaluated in the project. It is an ensemble learning method that combines multiple decision trees to produce a prediction.

The recorded test results were:

- MAE: 16.54 portions
- RMSE: 21.21 portions
- R^2 : 0.968

The model was able to produce predictions for the food-consumption target, although its recorded test metrics were different from those of the other two evaluated models.

expected_students	menu	day_of_week	weather	temperature_c	is_holiday	is_exam_day	special_event	Actual Consumption	Predicted Consumption	Prediction Error
135	Idli Sambar	Thursday	Normal	27.9	1	0	0	119	113.29	5.71
507	Rajma Rice	Thursday	Hot	36.6	0	0	0	440	444.53	-4.53
447	Dal Roti	Tuesday	Normal	30.4	0	1	0	398	399.26	-1.26
346	Idli Sambar	Saturday	Hot	36.6	0	0	0	281	289.48	-8.48
364	Idli Sambar	Saturday	Normal	31.2	0	0	0	323	299.85	23.15
370	Veg Pulao	Saturday	Normal	30.8	0	0	0	270	321.16	-51.16
431	Rajma Rice	Thursday	Cool	21.0	0	1	0	375	374.15	0.85
194	Pav Bhaji	Monday	Cloudy	30.4	1	0	0	172	180.02	-8.02
501	Dal Rice	Tuesday	Hot	33.5	0	0	0	418	436.50	-18.50
481	Paneer Rice	Thursday	Cloudy	30.8	0	0	0	457	420.39	36.61
460	Poha	Wednesday	Cool	19.4	0	1	0	352	391.08	-39.08
544	Dal Rice	Friday	Normal	26.5	0	0	0	442	474.74	-32.74
523	Veg Pulao	Friday	Normal	28.7	0	0	0	442	462.74	-20.74
211	Dal Roti	Sunday	Cool	25.6	0	0	0	211	176.33	34.67
397	Chole Bhature	Thursday	Normal	30.8	0	1	0	353	358.92	-5.92
182	Dal Rice	Monday	Hot	36.3	1	0	0	164	145.97	18.03
...										
488	Dal Roti	Wednesday	Normal	32.0	0	0	0	418	413.61	4.39
505	Rajma Rice	Friday	Normal	23.7	0	0	0	431	449.54	-18.54
222	Rajma Rice	Sunday	Normal	30.4	0	0	0	185	191.35	-6.35
415	Khichdi	Monday	Rainy	31.0	0	1	0	324	370.83	-46.83

Figure 3.3: Random Forest Regression model evaluation output.

Description:

This screenshot displays the Random Forest Regression model output and its evaluation metrics on the test dataset.

3.7 Model Performance Comparison

The three models were compared using MAE, RMSE, and R^2 .

Table 3.1: Model Performance Comparison

Model	MAE	RMSE	R^2
Linear Regression	14.11	16.84	0.980
Gradient Boosting	16.06	19.53	0.973
Random Forest	16.54	21.21	0.968

Meaning of Evaluation Metrics

1.MAE (Mean Absolute Error):

MAE represents the average absolute difference between the actual and predicted food consumption. A lower MAE indicates smaller average prediction errors.

2.RMSE (Root Mean Squared Error):

RMSE gives more importance to larger prediction errors. A lower RMSE indicates better control of larger errors.

R² (R-squared):

R² indicates how much of the variation in the target variable is explained by the model on the evaluated data.

3.8 Model Selection

Based on the recorded test results, Linear Regression was selected for the project's prediction workflow.

It achieved:

- MAE = 14.11 portions
- RMSE = 16.84 portions
- R² = 0.980

It also provides a relatively simple and interpretable model structure, which makes the prediction process easier to explain.

The selection was based on the recorded evaluation results rather than simply choosing a more complex algorithm.

3.9 Cross-Validation

In addition to the train-test evaluation, 5-fold cross-validation was used as an additional consistency check.

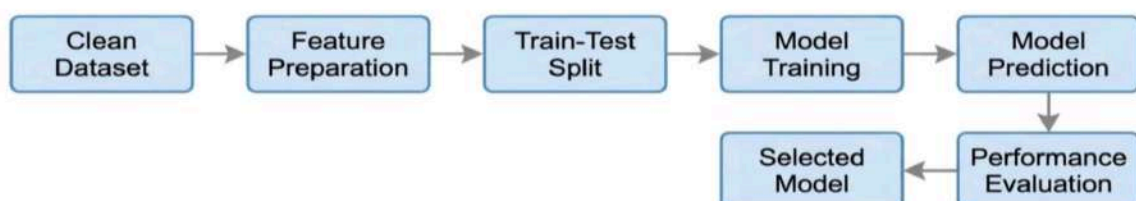
The recorded results were:

- Average MAE: 12.70 portions
- Standard deviation: 0.95

Cross-validation evaluates the model across multiple data partitions and provides additional information about the consistency of the model's performance.

3.10 Model Prediction Flow

The machine learning process follows these steps:



The selected model can then be used by the Prediction Engine to estimate food consumption for new conditions.

The selected model can then be used by the Prediction Engine to estimate food consumption for new conditions.

3.11 Chapter Summary

In this chapter, three machine learning regression models were evaluated for predicting food consumption: Linear Regression, Gradient Boosting Regression, and Random Forest Regression.

The dataset was divided into 240 training records and 60 testing records. The models were evaluated using MAE, RMSE, and R^2 .

Based on the recorded test results, Linear Regression achieved an MAE of 14.11, RMSE of 16.84, and R^2 of 0.980. Therefore, Linear Regression was selected for the project's prediction workflow.

The trained model provides the required prediction output for the next stage of the system, which is the Prediction Engine.

CHAPTER 4: PREDICTION ENGINE

4.1 Introduction

The Prediction Engine is an important component of the Food Waste Prediction and Optimization System. Its main purpose is to use the trained machine learning model to predict the amount of food that is likely to be consumed under given conditions.

Instead of preparing food only according to the expected number of students, the system considers different operational conditions such as the number of students, menu, day, weather, temperature, holiday status, examination status, and special events.

The predicted consumption is then used by the Recommendation Engine to calculate an appropriate food-preparation quantity.

4.2 Purpose of Prediction Engine

The main purposes of the Prediction Engine are:

- To predict expected food consumption.
- To use historical data for making predictions.
- To consider different operational conditions.
- To provide a numerical prediction in portions.
- To provide input for the Recommendation Engine.
- To support better food-preparation decisions.

The Prediction Engine does not directly determine the final preparation quantity. It first predicts the expected consumption, which is then passed to the Recommendation Engine.

4.3 Input Parameters

The Prediction Engine accepts the conditions that may affect food consumption.

The main input parameters are:

Input Parameter	Purpose
Expected Students	Number of students expected to be served
Menu	Food item being prepared
Day	Day of the week
Weather	Current weather category
Temperature	Temperature in °C
Holiday	Indicates whether it is a holiday
Exam Day	Indicates whether it is an examination day
Special	Indicates whether a special event is occurring

These inputs are processed in the same general feature structure used during model training.

4.4 Prediction Process

The prediction process follows a sequence of steps.

Step 1: Enter Input Conditions

The user provides the expected number of students and other relevant conditions.

Step 2: Feature Preparation

The entered categorical and numerical values are converted into the required format for the trained model.

Step 3: Apply Trained Model

The prepared input is passed to the selected machine learning model.

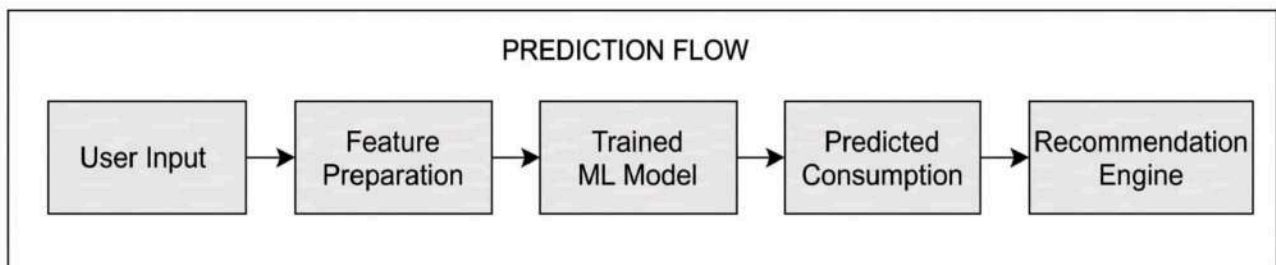
Step 4: Generate Prediction

The model generates the expected food consumption in portions.

Step 5: Display Result

The predicted consumption is displayed to the user and can then be used by the Recommendation Engine.

Prediction Flow:



4.5 Example Prediction

For the main demonstration scenario, the system receives approximately 500 expected students along with the selected menu and other operating conditions.

The trained model predicts approximately:

Predicted Food Consumption = 388 portions

This means that the model estimates that approximately 388 portions may be consumed under the selected conditions.

The prediction is then passed to the Recommendation Engine, where a safety buffer is applied before calculating the final recommended preparation quantity.

4.6 Prediction Interface

The project provides a user interface through which the user can enter the required conditions and obtain a food-consumption prediction.

The interface allows the user to provide information such as:

- Expected students
- Menu
- Day
- Weather
- Temperature
- Holiday status
- Examination status
- Special-event status

After entering the required values, the system displays the predicted food consumption.

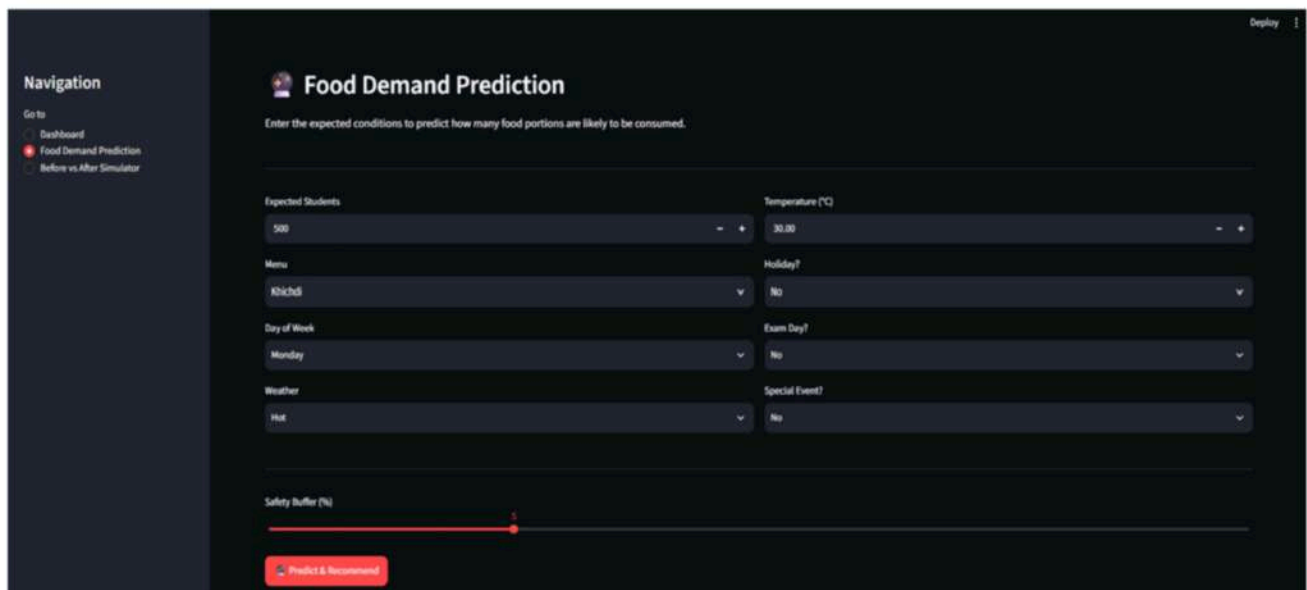
The screenshot shows a web application titled "Food Demand Prediction". On the left is a dark sidebar with a "Navigation" menu containing "Go to", "Dashboard", "Food Demand Prediction" (highlighted with a red dot), and "Before vs After Simulator". The main content area has a header "Food Demand Prediction" and a subtitle "Enter the expected conditions to predict how many food portions are likely to be consumed." Below this are several input fields: "Expected Students" (a numeric input with a value of 500), "Temperature (°C)" (a numeric input with a value of 30.00), "Menu" (a dropdown menu with "Khichdi" selected), "Holiday?" (a dropdown menu with "No" selected), "Day of Week" (a dropdown menu with "Monday" selected), "Exam Day?" (a dropdown menu with "No" selected), "Weather" (a dropdown menu with "Hot" selected), and "Special Event?" (a dropdown menu with "No" selected). At the bottom of the form is a "Safety Buffer (%)" slider set to 0 and a red "Predict & Recommend" button. A "Deploy" link is visible in the top right corner.

Figure 4.1: Food demand prediction interface of the Food Waste Prediction and Optimization System.

Description:

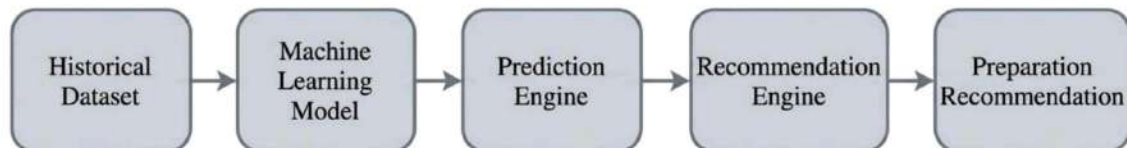
This screenshot shows the prediction interface where the user enters the expected number of students, menu, day, weather, temperature, and special-condition information. The system then displays the predicted food consumption in portions.

4.7 Role of Prediction in the Overall System

The Prediction Engine acts as a connection between the machine learning model and the Recommendation Engine.

The machine learning model provides the estimated consumption, while the Recommendation Engine uses this prediction to determine the recommended preparation quantity.

The overall process is:



This separation makes the system easier to understand and allows prediction and decision logic to be handled independently.

4.8 Benefits of Prediction Engine

The Prediction Engine provides the following benefits:

1. Data-driven prediction: Uses historical data instead of relying only on assumptions.
2. Condition-based prediction: Considers menu, day, weather, temperature, and special conditions.
3. Reduced over-preparation: Helps avoid preparing food only for the maximum expected attendance.
4. Decision support: Provides information for calculating the recommended preparation quantity.
5. Interactive use: Allows users to enter conditions and obtain predictions through the application.

4.9 Chapter Summary

The Prediction Engine uses the trained machine learning model to estimate food consumption for new operating conditions. It accepts information such as expected students, menu, day, weather, temperature, holiday status, examination status, and special events.

For the project's main demonstration, approximately 500 expected students resulted in a predicted consumption of 388 portions. This prediction is then passed to the Recommendation Engine, where a safety buffer is applied to determine the recommended preparation quantity.

Thus, the Prediction Engine provides the central prediction step required for the food-preparation decision-support process.

CHAPTER 5: RECOMMENDATION ENGINE

5.1 Introduction

The Recommendation Engine is an important module of the Food Waste Prediction And Optimization. Its main purpose is to convert the predicted food consumption into a practical food preparation recommendation.

Instead of preparing food only based on the total number of expected students, the system uses the predicted consumption value generated by the Machine Learning model. A small safety buffer is then added to handle unexpected variations in demand.

This approach helps avoid excessive food preparation while ensuring that sufficient food is available.

5.2 Purpose of Recommendation Engine

The main purposes of the Recommendation Engine are:

- To use the predicted food consumption value.
- To calculate a suitable safety buffer.
- To determine the recommended quantity of food to prepare.
- To reduce unnecessary food preparation.
- To support better food and cost management.
- To provide an easy-to-understand recommendation to the user.

The recommendation can be represented as:

Recommended Food Quantity = Predicted Consumption + Safety Buffer

5.3 Recommended Food Quantity

The Recommendation Engine takes the consumption predicted by the Machine Learning model as its main input.

For example, if the Prediction Engine estimates that approximately 388 portions will be consumed, the system does not simply recommend preparing exactly 388 portions. A small additional quantity is included as a safety buffer.

This provides flexibility when the actual demand is slightly higher than the prediction.

5.4 Safety Buffer

The safety buffer is an additional quantity added to the predicted consumption. It helps manage small differences between predicted and actual food demand.

In the project demonstration, a **5%** safety buffer is used.

For a predicted consumption of 388 portions:

$$\text{Safety Buffer} = 388 \times 5\%$$

$$\text{Safety Buffer} \approx 19 \text{ portions}$$

Therefore:

$$\text{Recommended Quantity} = 388 + 19$$

$$\text{Recommended Quantity} = 407 \text{ portions}$$

The safety buffer can be adjusted according to the requirements of the food preparation environment.

5.5 Example Recommendation

Consider a situation where the system receives the following prediction:

Parameter	Value
Expected Students	500
Predicted Consumption	388 portions
Safety Buffer	5%
Safety Buffer Quantity	19 portions
Recommended Preparation	407 portions

Thus, instead of preparing food for all 500 expected students, the system recommends approximately 407 portions based on the predicted consumption and safety buffer.

This recommendation is used as a project demonstration and should be interpreted as a simulation-based result.

5.6 Recommendation Process

The Recommendation Engine follows a simple sequence:

1. The user provides the required input information.
2. The Prediction Engine predicts food consumption.
3. The predicted consumption is received by the Recommendation Engine.
4. A predefined safety buffer is calculated.
5. The safety buffer is added to the predicted consumption.
6. The final recommended food quantity is calculated.
7. The recommendation is displayed to the user.

Recommendation Flow:

This process connects the Machine Learning model with the practical food preparation decision.

5.7 Recommendation Interface

The Streamlit application provides an interface for displaying the recommendation. The user can view the predicted consumption, safety buffer, and final recommended preparation quantity.

Description: The Recommendation Engine Interface Should show the recommendation result, including the predicted food consumption, safety buffer, and final recommended preparation quantity.

5.8 Benefits of Recommendation Engine

The Recommendation Engine provides several benefits:

- Helps reduce unnecessary food preparation.
- Converts machine learning predictions into practical decisions.
- Provides a safety margin for unexpected demand.
- Supports better food resource management.
- Can help reduce food wastage and associated costs.
- Provides a simple recommendation that can be easily understood by users.

5.9 Chapter Summary

The Recommendation Engine uses the predicted food consumption to calculate a practical preparation quantity. A safety buffer is added to account for small variations in actual demand. In the project demonstration, a prediction of 388 portions with a 5% safety buffer results in a recommendation of approximately 407 portions.

Thus, the Recommendation Engine acts as a bridge between machine learning predictions and practical food preparation decisions. It provides a simple and data-driven method for planning the required quantity of food.

CHAPTER 6: SYSTEM INTEGRATION AND DEPLOYMENT

6.1 Introduction

System integration is the process of combining the different modules of the Food Waste Prediction and Optimization System into a single working application. The project integrates data preparation, exploratory data analysis, machine learning, prediction, recommendation, and simulation into one application.

The system uses a Streamlit-based interface to provide users with an interactive way to access the different functions of the project. The integrated application allows the user to analyze food-waste information, provide prediction inputs, obtain food-consumption predictions, calculate recommendations, and compare preparation approaches.

The integration of these components provides a complete workflow from historical data analysis to practical food-preparation recommendations.

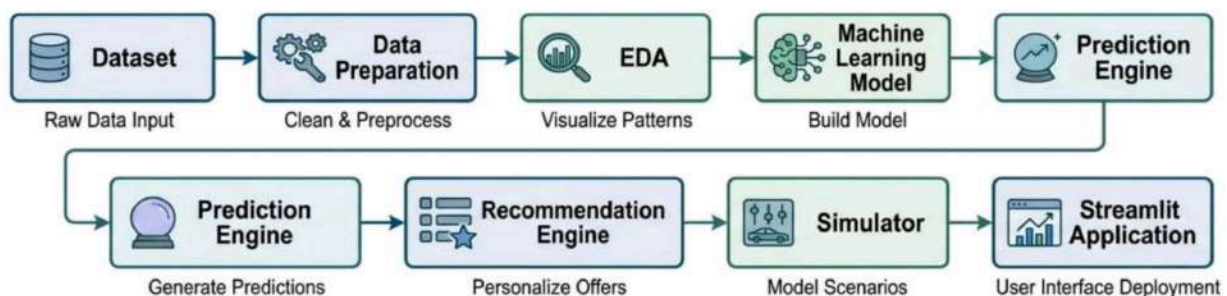
6.2 System Integration

The main components integrated into the system are:

- Dataset and data-preprocessing module
- Exploratory Data Analysis module
- Machine Learning model
- Prediction Engine
- Recommendation Engine
- Before-vs-After Simulator
- Streamlit user interface

These components work together to provide the complete functionality of the Food Waste Prediction and Optimization System.

System Integration Flow:



The data-preparation stage provides clean data for analysis and model development. The trained machine-learning model is then used by the Prediction Engine. The prediction is passed to the Recommendation Engine, where the safety buffer is applied. The Simulator uses these results to demonstrate the difference between traditional and AI-assisted food preparation.

6.3 Integration of Data Preparation and EDA

The cleaned dataset prepared in Chapter 1 is used as the foundation for the EDA module.

The EDA module uses the cleaned records to generate information about:

- Food preparation
- Food consumption
- Food waste
- Menu-wise waste
- Day-wise waste
- Monthly waste trends
- Weather-wise waste
- Special conditions

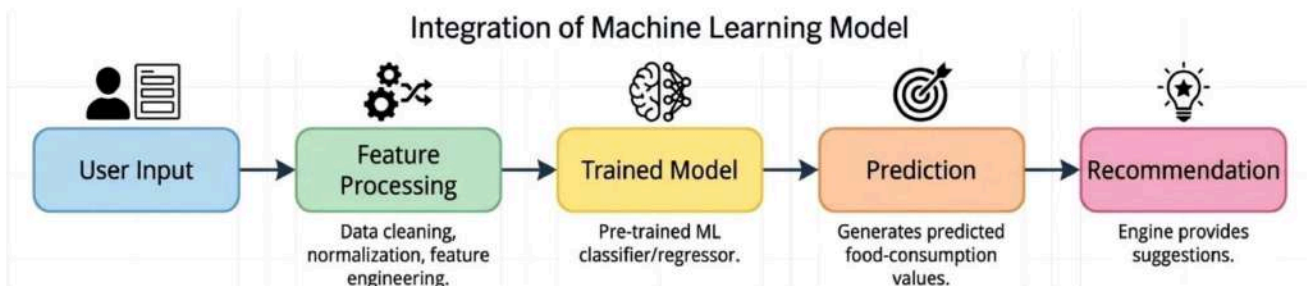
This allows the same prepared dataset to be used consistently throughout the analytical part of the application.

6.4 Integration of Machine Learning Model

The machine-learning component is integrated with the prediction functionality of the application.

The trained model receives the required input features and produces a predicted food-consumption value. The prediction is then made available to the Recommendation Engine.

The integration can be represented as:



This allows the machine-learning model to be used as part of the application's decision-support workflow rather than as an independent analysis program.

6.5 Integration of Prediction and Recommendation

The Prediction Engine and Recommendation Engine are connected sequentially.

First, the Prediction Engine estimates the expected food consumption. The Recommendation Engine then uses this prediction and applies the selected safety buffer.

For example:

Predicted Consumption = 388 portions

Safety Buffer = 5%

Safety Buffer Quantity $\approx$ 19 portions

Recommended Preparation $\approx$ 407 portions

This integration converts the numerical prediction into a practical preparation recommendation.

6.6 Integration of Before-vs-After Simulator

The Before-vs-After Simulator uses the prediction and recommendation results to demonstrate the potential effect of the proposed system.

The simulator compares:

Traditional Approach $\rightarrow$ Preparation based on expected students
with

AI-Assisted Approach $\rightarrow$ Preparation based on predicted consumption + safety buffer

For the project demonstration, the simulator can compare the traditional preparation quantity with the recommended quantity generated by the system.

The simulator therefore provides a simple way to understand the possible difference in food preparation and estimated waste.

6.7 Streamlit Application

The different project components are integrated through a Streamlit-based application.

The application provides an interactive interface through which users can access the functionality of the system without directly interacting with the underlying Python implementation.

The main application functionality includes:

- Dashboard and data analysis
- Food-consumption prediction
- Food-preparation recommendation
- Before-vs-After simulation

- Visualization of project results

The Streamlit interface acts as the front end of the project and connects the user with the underlying data-processing and machine-learning components.

6.8 Application Workflow

The complete application workflow is:

Step 1: Load Data

The system loads the prepared food-waste dataset.

Step 2: Analyze Data

The EDA module provides visualizations and statistical information about food waste.

Step 3: Enter Prediction Conditions

The user provides information such as expected students, menu, day, weather, temperature, and special conditions.

Step 4: Generate Prediction

The trained machine-learning model predicts the expected food consumption.

Step 5: Generate Recommendation

The Recommendation Engine applies the safety buffer and calculates the recommended preparation quantity.

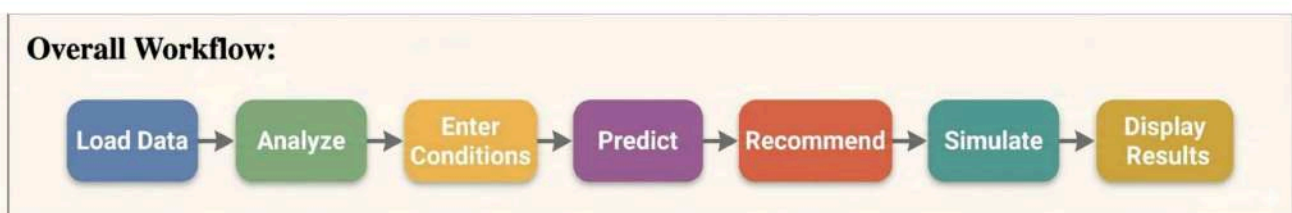
Step 6: Simulate Impact

The Before-vs-After Simulator compares the traditional and AI-assisted approaches.

Step 7: Display Results

The results are displayed through the Streamlit interface.

Overall Workflow:



6.9 Deployment

Deployment is the process of making the application available for use through its Streamlit interface.

The application can be executed in a Python environment containing the required project dependencies. The project files, dataset, machine-learning components, and Streamlit application are kept together so that the different modules can work as one system.

A typical deployment workflow is:



The deployment environment must contain the required Python libraries and project files used by the application.

Since the project is an academic prototype, deployment demonstrates the working application rather than a production-scale institutional deployment.

6.10 Application Requirements

The application requires a suitable Python environment and the libraries used by the project.

The basic requirements include:

- Python
- Streamlit
- Pandas
- NumPy
- Scikit-learn
- Matplotlib/visualization libraries used by the project
- Project dataset
- Project Python source files

The exact dependencies should match the libraries used in the project's implementation.

6.11 Testing of Integrated System

After integrating the different components, the system functionality can be checked through individual and end-to-end operations.

The important functions tested include:

Test Function	Expected Result
Dataset Loading	Dataset loads successfully
Data Validation	Invalid records are identified
EDA	Graphs and analysis are generated
Model Prediction	Food consumption is predicted
Recommendation	Recommended quantity is calculated
Simulator	Traditional and AI-assisted results are compared
Streamlit Interface	Application displays the required functionality

Testing helps ensure that the individual modules work correctly after being combined into the complete application.

6.12 Benefits of System Integration

The integrated system provides the following benefits:

1. **Single Application:** Different project functions are available through one interface.
2. **Connected Workflow:** Prediction and recommendation are directly connected.
3. **Interactive Analysis:** Users can interact with the application and view results.
4. **Decision Support:** Machine-learning predictions are converted into practical recommendations.
5. **Easy Demonstration:** The complete project can be demonstrated through the Streamlit application.
6. **Modular Structure:** Individual components can be improved or extended independently.

6.13 Chapter Summary

This chapter described the integration and deployment of the Food Waste Prediction and Optimization System. The project combines data preparation, EDA, machine learning, prediction, recommendation, and simulation into a single Streamlit-based application.

The integrated workflow begins with the prepared dataset and continues through analysis, prediction, recommendation, and simulation. The Streamlit interface provides users with access to these functions through an interactive application.

The system can be executed in an appropriate Python environment with the required dependencies and project files. As an academic prototype, the deployment demonstrates the integration and working of the project's different modules and provides a foundation for future expansion into a larger real-world food-management system.

CHAPTER 7 – STREAMLIT APPLICATION

7.1 Introduction

The Food Waste Prediction and Optimization system is implemented as an interactive web application using Streamlit. Streamlit provides a simple interface for accessing the different functionalities of the developed system.

The application acts as the user interface layer and connects the previously developed modules with the user. It allows users to provide the required inputs and view the generated results in an organized manner.

7.2 Streamlit Application Structure

The Streamlit application is designed to provide access to the major components of the Food Waste Prediction and Optimization system.

The main components available through the application are:

- Data analysis and visualization
- Food consumption prediction
- Food quantity recommendation
- Before-vs-After simulation

These components are integrated into the application so that the user can interact with the system without directly executing individual Python programs.

7.3 User Input

The application provides interactive input controls for entering the parameters required by the prediction model.

The main input parameters are:

- Expected number of students
- Menu item
- Day of the week
- Weather condition
- Temperature
- Holiday status
- Examination-day status

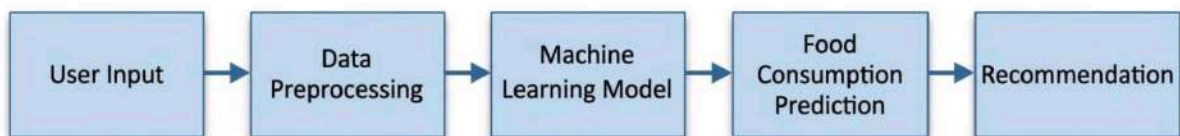
- Special-event status

These values are processed by the application and passed to the prediction workflow.

7.4 Application Processing

After the user provides the required inputs, the Streamlit application processes the information through the developed prediction workflow.

The general processing sequence is:



The predicted consumption is then used to calculate the recommended preparation quantity with the selected safety buffer.

For the demonstrated example:

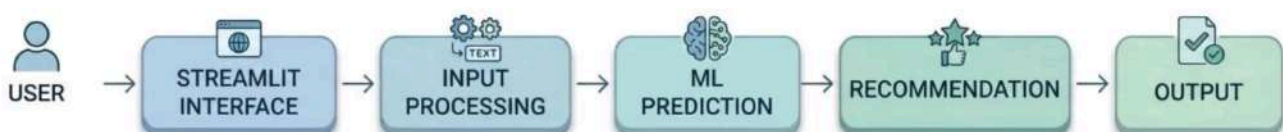
- Predicted consumption = 388 portions
- Safety buffer = 5%
- Safety buffer quantity $\approx$ 19 portions
- Recommended preparation quantity $\approx$ 407 portions

7.5 Application Output

The results generated by the system are displayed directly on the Streamlit interface. The output provides the user with the predicted food consumption and the recommended preparation quantity. The application can also display analytical graphs and simulation results when the corresponding functionality is selected.

This provides a convenient way to understand the results without examining the underlying Python code.

7.6 Application Workflow



The application connects the user interface with the different modules developed during the project. This makes the complete Food Waste Prediction and Optimization workflow accessible through a single interface.

7.7 Application Testing

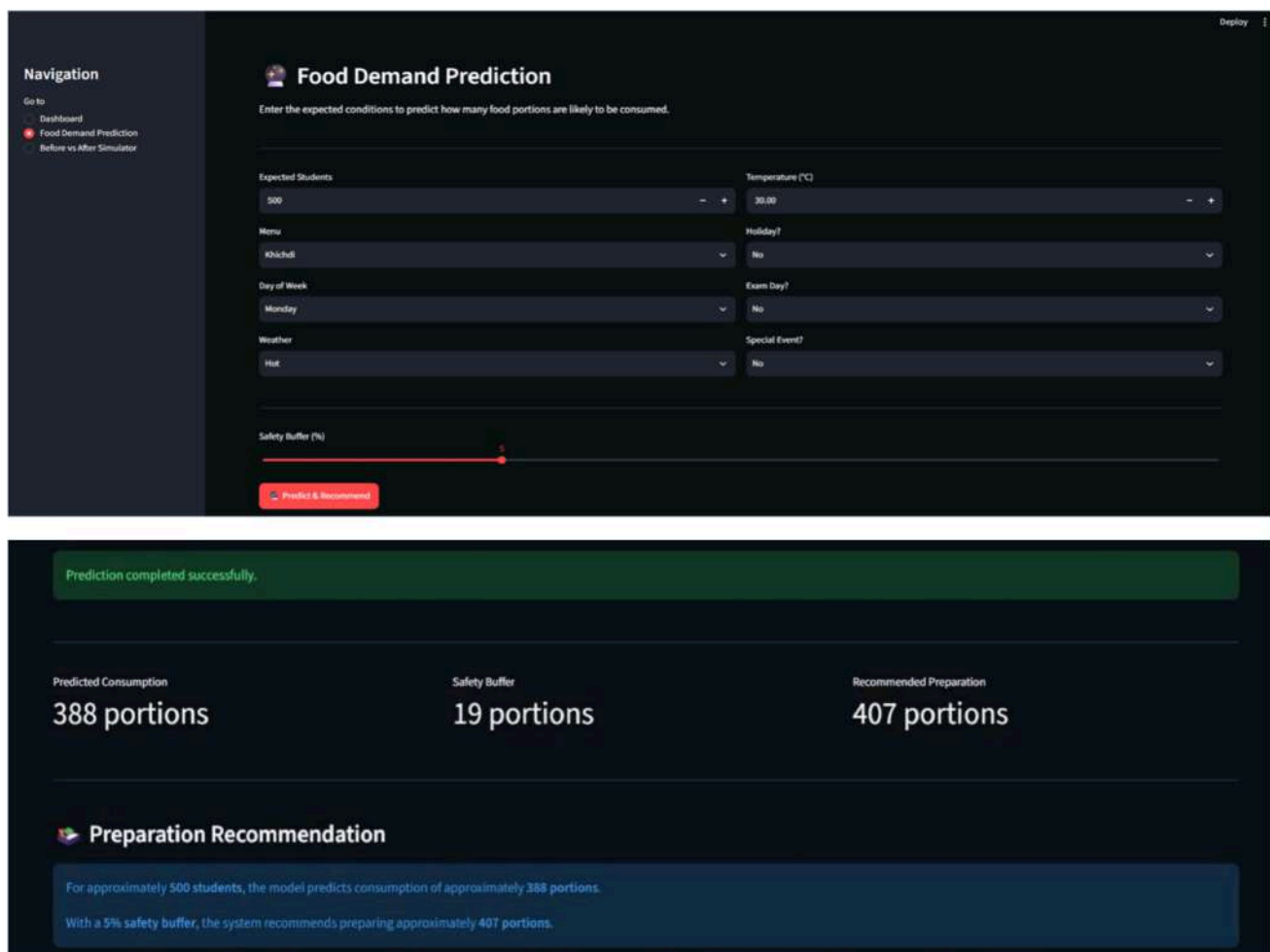
The Streamlit application was tested to verify the operation of the major functions.

Test Component	Expected Result
Application Loading	Interface loads successfully
User Input	Input values are accepted correctly
Prediction	Food consumption prediction is generated
Recommendation	Recommended quantity is calculated
Data Visualization	Required graphs are displayed
Simulation	Comparison results are generated

The testing confirmed that the major application functions operate through the integrated interface.

7.8 Screenshot of Streamlit Application

Figure 7.1: Food Waste Prediction and Optimization – Streamlit Application



Description:

The screenshot shows the working Streamlit interface of the Food Waste Prediction and Optimization system. It demonstrates the user input section and the corresponding system output in the application.

7.9 Chapter Summary

The Streamlit application provides the interactive interface for the Food Waste Prediction and Optimization system. It connects the machine learning model, prediction workflow, recommendation process, data visualization, and simulation functionality through a single user interface.

The application simplifies interaction with the developed system and provides the generated results in an organized and user-friendly manner.

13. RESULTS & CONCLUSION

Results

The Food Waste Prediction and Optimization system was successfully developed by combining data preparation, Exploratory Data Analysis, machine learning, prediction, recommendation, simulation, and Streamlit application modules.

The dataset contained 300 records and 18 columns. During data validation, 5 invalid records were identified where the number of food portions consumed was greater than the number of actual students. After correction, the validation condition returned no remaining invalid records.

The analysis of the dataset produced the following overall results:

Parameter	Result
Average Food Prepared	403.07 portions
Average Food Consumed	340.32 portions
Average Food Wasted	62.75 portions
Average Waste Percentage	15.83%
Total Food Wasted	18,824 portions

Three machine learning regression models were evaluated for predicting food consumption.

Model	MAE	RMSE	R ²
Linear Regression	14.11	16.84	0.980
Gradient Boosting Regression	16.06	19.53	0.973
Random Forest Regression	16.54	21.21	0.968

Based on the recorded test results, Linear Regression was selected for the prediction workflow. The model achieved an R² value of 0.980, indicating a strong relationship between the selected input features and predicted food consumption.

The 5-fold cross-validation produced an average MAE of 12.70 with a standard deviation of 0.95, providing an additional evaluation of model performance.

For the demonstration prediction, the system estimated food consumption of 388 portions for an input of 500 expected students. With a 5% safety buffer of approximately 19 portions, the recommended preparation quantity was approximately 407 portions.

The Before-vs-After Simulator demonstrated a comparison between traditional preparation and AI-assisted preparation. In the example simulation, preparation was reduced from 500 portions to 407 portions, with estimated waste changing from 112 portions to 19 portions. These values are simulation-based and do not represent measured real-world savings.

Conclusion

The Food Waste Prediction and Optimization project demonstrates how data analysis and machine learning can be combined to support food preparation decisions. The system analyzes historical food consumption and waste information, identifies important patterns, predicts expected food consumption, and generates a recommended preparation quantity.

The developed machine learning models were evaluated using MAE, RMSE, and R^2 metrics, with Linear Regression providing the recorded best test performance among the evaluated models. The Prediction Engine and Recommendation Engine then use the model output to support preparation decisions with an additional safety buffer.

The Streamlit application provides an interactive interface for accessing the developed system and makes the prediction, recommendation, analysis, and simulation functions easier to use.

Overall, the project successfully demonstrates a complete Food Waste Prediction and Optimization workflow. Since the dataset used in the project is custom/synthetic, the reported reduction in waste from the simulator should be considered a demonstration of the system's potential rather than measured real-world performance.

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15. APPENDIX — PROJECT STRUCTURE

```
|— app.py
|— food_waste_prediction_optimization_dataset_cleaned.csv
|
|— src/
    |— prediction_engine.py
    |— recommendation_engine.py
    |— simulator.py
    |— ml_model.py
    |— eda.py
```

Description of Project Files

[app.py](#)

Main Streamlit application that integrates the different modules of the Food Waste Prediction and Optimization system.

[food_waste_prediction_optimization_dataset_cleaned.csv](#)

Cleaned dataset used for data analysis, machine learning, prediction, and optimization.

[src/prediction_engine.py](#)

Contains the functionality for predicting food consumption using the trained machine learning model.

[src/recommendation_engine.py](#)

Calculates the recommended food preparation quantity using the predicted consumption and safety buffer.

[src/simulator.py](#)

Implements the Before-vs-After Simulator for comparing traditional and AI-assisted food preparation.

[src/ml_model.py](#)

Contains the machine learning model implementation and model-related processing

[src/eda.py](#)

Contains the Exploratory Data Analysis functionality, including analysis and visualization of the food-waste dataset.